communications earth & environment

Article

A Nature Portfolio journal

https://doi.org/10.1038/s43247-026-03461-8

Recent cloud trends and extremes reaffirm established bounds on cloud feedback and aerosol-cloud interactions

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Earth's energy imbalance has increased markedly over the past two decades, reaching a record in 2023. Both the long-term trend and year-to-year variations are linked to reduced reflection of sunlight by low-level clouds, which is pronounced over Northern Hemisphere oceans. However, the causes of these cloud changes and their implications for future Earth system evolution are unknown, raising concerns about a stronger-than-expected cloud feedback. Here we quantify the meteorological factors behind interannual cloud-radiative anomalies, several of which aligned to produce the extreme 2023 value. These meteorological variations are superposed on a background of declining sulfate aerosol concentrations, contributing to a sustained decrease in cloud reflection over the past 22 years. The resulting constraints on cloud feedback and aerosol forcing are consistent with previous studies, supporting an equilibrium climate sensitivity near 3°C (likely range 2.7–4.1°C). Thus, recent observations do not indicate an emerging stronger cloud feedback or underestimated future warming.

Recent studies reveal a marked increase in Earth's energy imbalance (EEI) over the past 25 years, driven by the steady decline in planetary albedo[1,2]. Notably, Earth's albedo reached a record minimum in 2023[3], coinciding with a large spike in global mean surface temperature[4]. Global climate models generally struggle to simulate EEI trends as large as observed, particularly in the shortwave[5–7]. This may implicate biases in model representation of key physics governing cloud feedback or aerosol-cloud interactions, forcing errors, or some combination of both.

A substantial portion of both the trend and the extreme albedo anomaly in 2023 stems from decreases in low-level cloud amount or reflectivity[1,8], but the exact reasons for these cloud changes remain unknown[3]. Decadal cloud trends and recent extremes result from internal climate variability and forced responses involving diverse drivers. Secular increases in greenhouse gases drive warming that may reduce cloud reflection through positive cloud feedbacks[9,10], while simultaneous decreases in aerosol concentrations[11–14] may reduce cloud reflection through aerosol-cloud interactions. Recent work has highlighted the potential role of new shipping fuel regulations in reducing sulfate aerosol loading[15–17], but the ensuing cloud-radiative impacts remain uncertain.

Recent trends and extremes in EEI, planetary albedo, and surface temperature, along with models' apparent inability to capture them, have prompted speculation that cloud feedbacks, aerosol-cloud interactions, or climate sensitivity are currently underestimated[3,18]. Here we show that observed trends and extreme cloud-induced radiation anomalies can be reproduced via "cloud-controlling factor" (CCF) analysis using observed sensitivities of cloud-radiative effects to meteorology and aerosols. Its success strengthens confidence in the CCF framework, while enabling attribution of cloud responses to specific controlling factors. We then show that observational constraints on cloud feedback and the effective radiative forcing from aerosol-cloud interactions (ERF_{aci}) derived using this updated unified set of CCF coefficients are consistent with previous studies and with a moderate equilibrium climate sensitivity (ECS), in agreement with existing consensus. Hence, we conclude that recent observations do not imply that larger-than-expected cloud feedback or ERF_{aci} are emerging in nature or that future warming has been underestimated.

Results and Discussion

Cloud-radiative sensitivity to controlling factors

We estimate the sensitivity of downwelling shortwave radiative anomalies induced by low-level clouds (R_{cld}) to a set of 7 commonly used low-cloud controlling factors[19–22]: Sea surface temperature (SST), estimated inversion strength (EIS), surface temperature advection (Tadv), 700 hPa relative

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humidity (RH_{700}), 700 hPa pressure velocity (ω_{700}), 10-meter wind speed (WS), and the natural logarithm of sulfate aerosol optical depth at 550 nm (*ln*(SAOD)). The first 6 (meteorological) CCFs come from European Centre for Medium-Range Weather Forecasts Reanalysis v5 (ERA5) reanalysis, while the 7th (sulfate aerosol) CCF comes from the Copernicus Atmosphere Monitoring Service (CAMS) reanalysis.

We first compute R_{cld} by convolving Moderate Resolution Imaging Spectroradiometer (MODIS) cloud fields[23] between 2003 and 2024 with cloud radiative kernels that quantify clouds' impact on the top-of-atmosphere radiation[24,25]. We then estimate coefficients quantifying the sensitivity of R_{cld} to each CCF within 5°x5° grid boxes across the global ocean. This is done using ordinary least-squares multilinear regression of detrended R_{cld} anomalies on detrended and standardized CCF anomalies. Further details on the CCF analysis, along with sensitivity tests related to alternative choices of aerosol predictors and cloud-radiative datasets are provided in *Methods*.

Maps of the local sensitivity of R_{cld} to each of the seven CCFs are shown in Supplementary Fig. 1, column 1. At most marine locations, the SST, Tadv, and ω_{700} coefficients are positive, indicating that increased surface temperature, decreased cold advection, and increased subsidence tend to favor fewer or optically thinner low-clouds and hence reduced albedo. Conversely, the EIS, RH_{700}, WS, and aerosol coefficients are negative at nearly every location, indicating that increased inversion strength, humidity above the boundary layer, surface wind speed, and aerosol loading lead to more or optically thicker low-clouds and hence increased albedo. These sensitivities are consistent with known meteorological controls on low cloud properties and with previous studies[19,21].

Anomalies and trends in controlling factors

Supplementary Fig. 1, columns 2 and 3 show CCF anomalies for 2023 and trends over 2003–2024, respectively. SSTs were anomalously warm nearly everywhere in 2023, particularly in the tropical Eastern and Northwest Pacific and tropical Atlantic, extending into northern midlatitudes. In these hot spots, SST anomalies exceeded 2 standard deviations and were accompanied by reduced EIS and cold advection. The extreme Atlantic SSTs in 2023 have been linked to anomalously weak winds that produced a shallow mixed layer highly sensitive to coincident positive radiative and turbulent heat flux anomalies[26]. The 22-year SST trend resembles the 2023 anomaly but shows minimal warming or slight cooling across the equatorial and Southeastern Pacific and subtropical North Atlantic. Positive EIS trends and negative RH_{700} trends appear across most ocean basins, particularly over the Central and Eastern Pacific where surface warming is suppressed. These widespread positive EIS trends contrast with the localized strong EIS reductions in 2023. Tadv trends are muted compared to 2023's regional anomalies. Vertical motion anomalies in 2023 indicate increased Central and Eastern Pacific ascent (consistent with El Niño), while the trend suggests overall circulation slowdown with decreased subsidence and ascent at 700 hPa. Large surface wind speed reductions over the Atlantic are evident in 2023, whereas the 22-year trend highlights decreased subtropical Pacific trade winds. Finally, large negative SAOD anomalies (exceeding 2 standard deviations in many locations) appear throughout the Northern Hemisphere in 2023, primarily reflecting the downward trend from reduced industrial emissions (including from shipping).

Causes of cloud radiative anomalies and trends

To predict R_{cld} anomalies in 2023, we multiply coefficient maps (Supplementary Fig. 1, column 1) by corresponding CCF anomaly maps (Supplementary Fig. 1, column 2). Maps showing these individual products are shown in Supplementary Fig. 2. Note that when estimating CCF coefficients, we use detrended data and iteratively exclude the target year and surrounding 5 years from the dataset used to fit the model (Methods); hence, R_{cld} anomalies reconstructed via CCF analysis in each year are out-of-sample tests. The sum of these products provides an estimate of the total R_{cld} response, which closely matches the actual 2023 anomaly in spatial structure and in magnitude, with a 60°S – 60°N pattern correlation of 0.77 (Fig. 1a-b). Large positive SW heating anomalies across the North Pacific, NE Atlantic, and tropical Eastern Pacific are well-reproduced, validating the CCF approach for capturing observed low-cloud induced radiative anomalies during this extreme year. The spatial structure of these R_{cld} anomalies primarily comes from the combined meteorological contributions, but reduced sulfate aerosol loading contributes uniformly positive anomalies throughout NH oceans (Fig. 1c, d).

As with the 2023 anomalies, trends in R_{cld} are closely reproduced by CCF analysis, showing large increases in the North Pacific and North Atlantic, especially off the coast of North America, and large decreases in the South Pacific, especially off the coast of South America (Fig. 1e, f). The 60°S – 60°N pattern correlation between reconstructed and actual R_{cld} trend maps is 0.69. The heterogeneous spatial pattern of these trends mainly arises from the meteorological trends, but the more widespread and homogeneous downward aerosol trends across the NH oceans make a substantial contribution that is uniformly positive (Fig. 1g, h).

Given the large regional R_{cld} signals over NH oceans, we now focus on the North Pacific and North Atlantic domains, where interannual low cloud radiative anomalies are closely reproduced by CCF analysis (Fig. 2a, c). Interannual variations in R_{cld} primarily stem from meteorological fluctuations (blue lines in Fig. 2b, d; shading in Fig. 2a, c). Notably, the large 2023 North Atlantic anomaly arises primarily from a pronounced spike in meteorological contributions (discussed below). The sulfate aerosol contribution to R_{cld} exhibits a positive secular trend over 2003–2024, such that a given positive meteorological contribution near the end of the record causes a larger positive R_{cld} anomaly than one near the beginning. Hence, the extreme 2023 North Atlantic cloud radiative anomalies arise from a remarkable alignment of cloud-unfavorable meteorological conditions superposed on a background of nearly monotonically decreasing aerosol loading.

R_{cld} anomalies in some years are poorly reconstructed via CCF analysis, notably 2011 and 2022 (Fig. 2c, d). These discrepancies may result from inaccuracies in derived CCF coefficients, errors in CCF or R_{cld} anomalies, cloud variations unaccounted for by the CCFs, or some combination of these. While the seven CCFs explain most of the R_{cld} variance, they cannot account for all of the variance at every marine location (Supplementary Fig. 4). Nevertheless, reconstructed and observed R_{cld} anomaly maps from 2003 to 2024 closely match throughout the global oceans (Supplementary Fig. 5). Area-weighted pattern correlation coefficients between annual reconstructed and observed R_{cld} anomaly maps over the 60°S-60°N oceanic domain range from 0.64 to 0.85, with a mean of 0.71.

Examining individual contributors to the 2023 R_{cld} anomalies in the North Pacific and Atlantic domains reveals that no CCFs make a substantive negative contribution (Fig. 3a, c). In the North Pacific, anomalously warm SSTs, low RH_{700}, and reduced sulfate aerosols lead to the positive cloud radiative anomaly that closely matches the observed anomaly in 2023. In the North Atlantic, all CCFs except RH_{700} and ω_{700} make substantive positive contributions. Notably, high SSTs combined with weak inversion strength, weak cold advection, and reduced aerosol loading yield cloud radiative anomalies approaching 3 W m^{-2}. This alignment of multiple meteorological and aerosol influences in 2023 strongly reduces planetary albedo — a configuration unique in the 22-year record. Quantifying the likelihood of such compound hostile influences on low-level clouds and whether it may change in response to radiative forcing is important future work.

Averaged over the North Pacific, increasing SST and decreasing SAOD are the dominant contributors to the positive R_{cld} trend, each making a ~ 0.3 W m^{-2} $decade^{-1}$ contribution (Fig. 3b). Smaller positive contributions from Tadv, RH_{700}, and WS are offset by small negative contributions from increased EIS. In the North Atlantic, meteorological CCF trends roughly cancel as strong EIS-induced R_{cld} decreases oppose smaller R_{cld} increases from rising SST and declining cold advection and RH_{700} (Fig. 3d). Hence the positive R_{cld} trend of ~ 0.4 W m^{-2} $decade^{-1}$ in the North Atlantic over 2003–2024 is almost entirely due to declining sulfate aerosol loading.

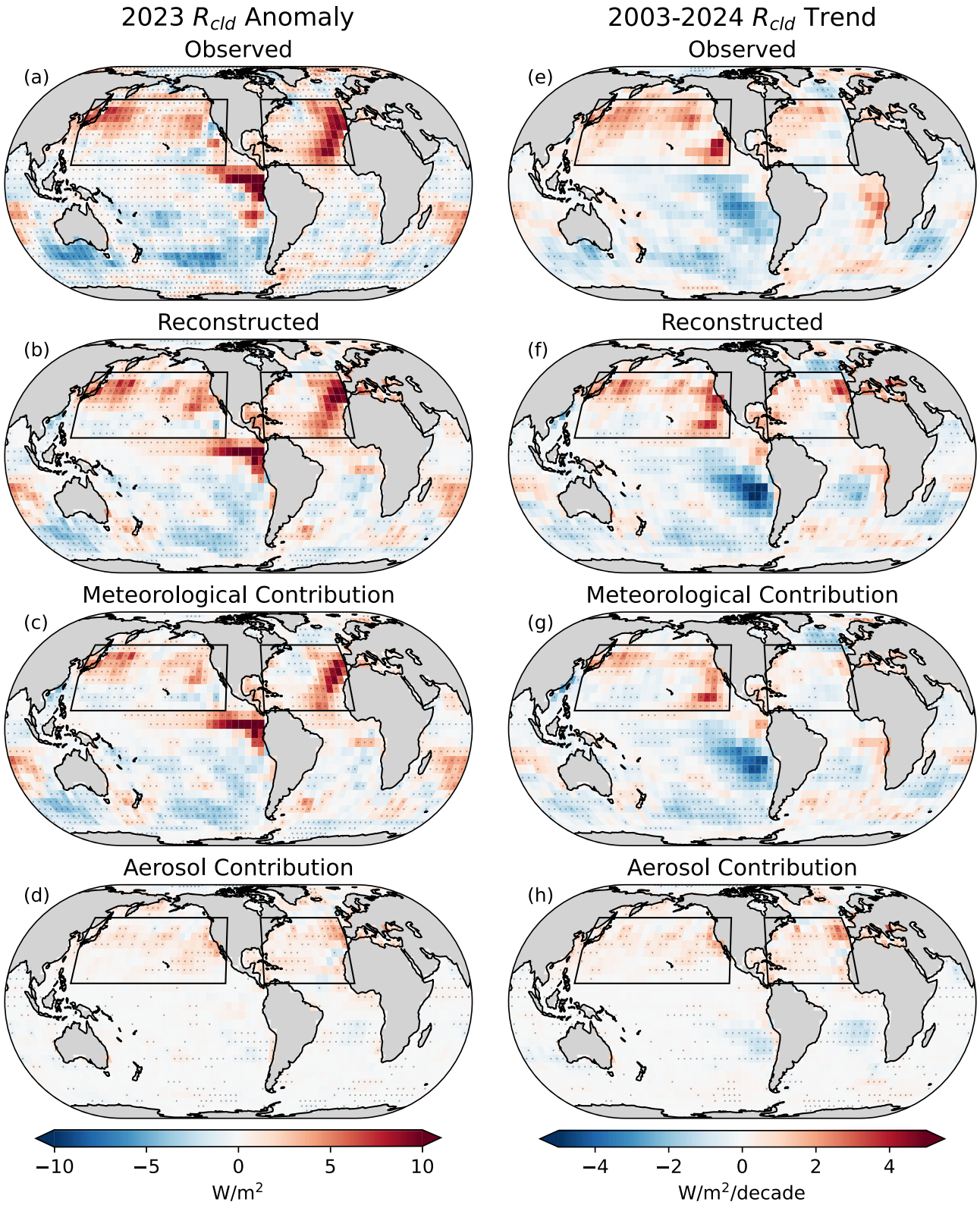


Fig. 1 | Observed and reconstructed anomalies and trends in low-cloud induced downwelling shortwave radiation anomalies (R_{cld}). (**column 1**) R_{cld} in 2023 as (**a**) observed and (**b**) reconstructed via cloud controlling factor analysis, along with contributions to these anomalies from (**c**) meteorology and (**d**) sulfate aerosols; their sum is equal to the reconstructed map in (**b**). (**column 2**) R_{cld} trends between 2003 and 2024 as (**e**) observed and (**f**) reconstructed via cloud controlling factor analysis, along with contributions to these anomalies from (**g**) meteorology and (**h**) sulfate aerosols; their sum is equal to the reconstructed map in (**f**). Meteorology represents the first six cloud controlling factors (i.e., all factors except sulfate aerosols). Stippling represents statistical significance at 95% confidence. Boxes enclose the North Pacific (10–50°N, 130–250°E) and North Atlantic (10-50°N, 280-350°E) regions analyzed further in Figs. 2 and 3.

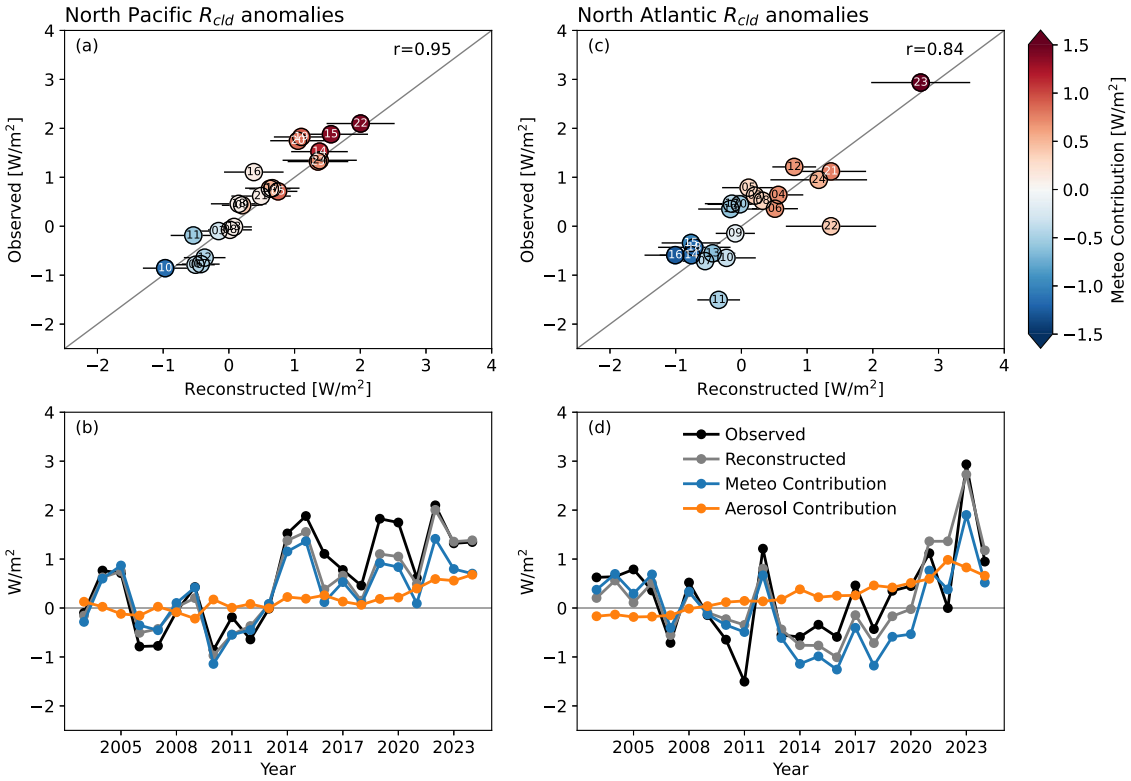


Fig. 2 | Observed versus reconstructed R_{cld} anomalies and their time evolution. R_{cld} anomalies in each year scattered against those reconstructed by CCF analysis, for the (**a**) North Pacific and (**c**) North Atlantic regions enclosed by boxes in Fig. 1. Each marker is color-coded by the meteorology contribution in that year. Errorbars indicate 95% confidence intervals. (**b**, **d**) show the temporal evolution of observed and reconstructed low-cloud-radiative anomalies, along with the contributions to them from meteorology and aerosols.

Implications for aerosol forcing, cloud feedback, and climate sensitivity

Two key points from the previous sections are that (1) both the extreme regional R_{cld} anomalies in 2023 and the 22-year R_{cld} trend over the MODIS record are closely reconstructed using observed sensitivities of R_{cld} to CCFs; and (2) these sensitivities are very similar to those used previously to observationally constrain low-cloud feedback (λ_{cld}) and ERF_{aci}[19–22]. It seems unlikely, therefore, that recent observed R_{cld} extremes and trends require a revision to our existing λ_{cld} and ERF_{aci} estimates. In this section we verify this quantitatively by repeating these analyses with the unified set of CCF coefficients derived herein. In what follows, we use net (LW+SW) CCF coefficients, which are nearly identical to the SW coefficients used above owing to the small LW effect of low-clouds.

We derive an observationally-constrained ERF_{aci} estimate by multiplying the observed sensitivity of R_{cld} to aerosols by the CMIP6 mean historical change in *ln*(SAOD) following ref. 21 (*Methods*; Supplementary Fig. 6). Our ERF_{aci} estimate of -1.20 ± 0.61 W m^{-2} (90% confidence) is very similar to that of ref. 21, whose methodology we closely follow (Fig. 4a). Our central estimate is larger in magnitude than that from IPCC AR6[27] and the WCRP assessment[28], but uncertainty ranges overlap considerably. However, our value significantly exceeds that of ref. 29, who derive weaker sensitivity of R_{cld} to *ln*(SAOD) by considering how this relationship is mediated through droplet number concentration (N_d). Accounting for N_d mediation in our analysis also yields weaker negative ERF_{aci} values, but full reconciliation with ref. 29 requires further investigation. ERF_{aci} estimates are sensitive to whether N_d mediation is considered and which observational datasets are used for N_d and the aerosol predictor (*Methods*).

We derive an observationally-constrained λ_{cld} estimate by multiplying the observed sensitivity of R_{cld} to the six meteorological CCFs by the modeled CCF changes under abrupt quadrupling of CO_2 (Supplementary Fig. 7a–f) following ref. 20 (*Methods*). These products and their sum are shown in Supplementary Fig. 7g–m. In response to quadrupled CO_2, the CCF that experiences the largest change relative to its interannual variability is SST, as also shown in several previous studies[20,22,30,31]. The sensitivity of low clouds to SST is largest in regions of marine stratocumulus, as shown in Fig. 1a and in several previous studies[19,22,32]. Therefore, the SST component of the cloud feedback - estimated as this large change in SST times the sensitivity of R_{cld} to SST - dominates the overall response of low clouds to CO_2-induced warming. Our resulting λ_{cld} estimate of $+0.21 \pm 0.23$ W m^{-2} K^{-1} (90% confidence) agrees closely with ref. 20, whose methods we follow most closely, but has larger uncertainty because we include ΔCCFs from 85 rather than 18 CMIP models (Fig. 4b). While our 66% ranges overlap considerably, our central estimate is lower than WCRP[10] and IPCC[27] assessments, which rely on more than just CCF analysis, and is lower than the analysis of ref. 22, which uses a different methodology that favors higher λ_{cld} despite qualitatively similar coefficient maps. Importantly, our coefficients yield an implied cloud feedback within previous uncertainty ranges, suggesting that recent observations do not indicate that a new stronger cloud feedback is emerging in nature.

Finally, we examine implications of our observationally-constrained ERF_{aci} and λ_{cld} values for ECS using the WCRP ECS Assessment methodology that combines paleoclimate, historical, and process understanding evidence in a Bayesian framework to produce a posterior ECS distribution[10]. Following refs. 20,21, we replace the expert-assessed ERF_{aci} and λ_{cld} values in the historical and process understanding lines of evidence with our derived values, while keeping paleoclimate evidence unchanged. The resulting central ECS estimate (3.3 °C) with a likely (66% confidence) range of 2.7–4.1 °C is shifted only slightly upward from that derived in the original WCRP assessment (Fig. 4c). This is because the lower ECS implied by our slightly weaker positive λ_{cld} is counteracted by a higher ECS implied by our slightly stronger negative present-day ERF_{aci} estimate (Supplementary Fig. 8). This posterior ECS shows some sensitivity to whether and how N_d mediation is handled (affecting historical ERF_{aci}), but minimal sensitivity to R_{cld} predictand choice (Supplementary Fig. 9). The likely ECS range remains

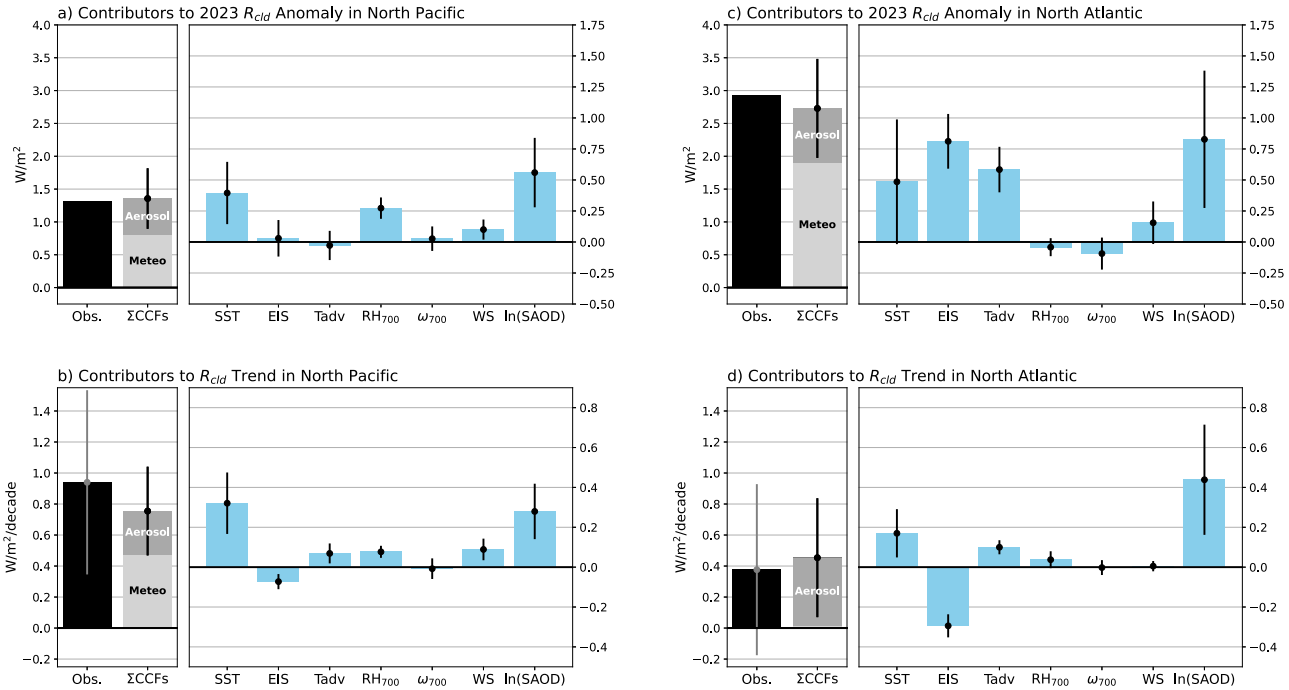


Fig. 3 | Individual CCF contributions to the 2023 anomalies and trends in R_{cld} in the North Pacific and North Atlantic domains. Observed ("Obs.") and reconstructed ("ΣCCFs") anomalies in R_{cld} in the (**a**) North Pacific and (**c**) North Atlantic regions. The reconstructed anomalies are shown as a stacked barplot, with the aerosol contribution stacked atop the sum of meteorological contributions. Positive values indicate increased absorbed shortwave radiation. Blue bars indicate the breakdown into CCF contributions (on a different vertical scale). (**b**, **d**) show the corresponding fields for the R_{cld} trends between 2003 and 2024. Error bars indicate 95% confidence intervals.

within 2.5–4.1 °C across sensitivity tests. Despite a larger central ECS estimate compared to IPCC AR6 (3.3 vs. 3.0 °C), our posterior likely range is roughly identical to those from both the WCRP and IPCC Assessments. Hence, recent observations affirm moderate ECS, with < 1.5% probability of ECS below 2 K.

Conclusions

Observationally-derived coefficients quantifying low-cloud sensitivity to meteorology and aerosols, multiplied by observed changes in these factors, successfully captures observed trends in low-cloud radiative effects over the past 22 years and recent record anomalies. These coefficients, when combined with model-predicted changes in meteorology and aerosols, yield observationally-constrained estimates of indirect aerosol forcing and low-cloud feedback within the range of previous studies, implying climate sensitivity near 3°C, consistent with community assessments. Contrary to the speculation put forward in several prominent studies, recent trends and extremes do not indicate stronger-than-expected aerosol forcing, cloud feedbacks, or climate sensitivity; rather they are consistent with known sensitivities of clouds to their environment. While cloud-radiative anomalies and trends over the past 22 years are within expectations, continued global monitoring of clouds and radiation remains crucial to assess how Earth's energy budget is impacted by the likely larger warming, continued aerosol reductions, and evolving surface temperature patterns of future years.

Methods

Predictor and Predictand Data

Our analysis focuses on cloud-induced shortwave (SW) radiative anomalies at the top-of-the-atmosphere (TOA) due to low-level clouds (R_{cld}). These anomalies are derived by convolving histograms of cloud fraction partitioned into bins of cloud top pressure and visible optical depth with SW cloud radiative kernels, which quantify the sensitivity of TOA SW radiative fluxes to unit perturbations in cloud fraction in these same bins[24,25]. Cloud fraction histograms are taken from the Moderate Resolution Imaging Spectroradiometer (MODIS) collection 6.1 dataset that has been produced to facilitate comparison with MODIS simulator output in climate models, known as MODIS-COSP[23]. This dataset is based on cloud retrievals from the MODIS instrument on board both Aqua and Terra satellites. The product of cloud fraction anomalies and cloud radiative kernels, when summed over the histogram bins corresponding to low-level clouds (i.e., cloud top pressures greater than 680 hPa), yields monthly resolved low-cloud-induced SW radiation anomalies over the duration of the MODIS record. These are then adjusted to remove radiation anomalies due to changes in obscuration by overlying clouds[19,20,33]. We similarly compute low-cloud induced net (LW + SW) radiation anomalies by convolving the cloud fraction anomalies with the sum of LW and SW cloud radiative kernels. Because the impact of low-level clouds on longwave radiation at the TOA is small, SW and net R_{cld} anomalies are very similar.

As a secondary cloud-radiative dataset, we also analyze the Clouds and the Earth's Radiant Energy System Flux-By-Cloud-Type (CERES-FBCT) product[34]. FBCT provides cloud fraction histograms as well as histograms of cloud radiative effects partitioned by cloud top pressure and optical depth. We compute the monthly-resolved climatological mean of this cloud radiative effect histogram – which is analogous to the cloud radiative kernel – and convolve it with the time-varying cloud fraction histograms, yielding cloud radiative effects due to each individual cloud type. Unobscured low-cloud radiative effects are then estimated in the same way as for the MODIS-COSP dataset. Two versions of the FBCT dataset are available: Edition 4A data is based on cloud and radiation retrievals from MODIS and CERES instruments on the Terra and Aqua satellites and is available from July 2002 to February 2023. Edition 1B data is based on cloud and radiation retrievals from Visible Infrared Imaging Radiometer Suite (VIIRS) and CERES instruments on the NOAA-20 satellite and is available from May 2018 to July 2024. After computing unobscured low-cloud radiative effects from both of these datasets, we concatenate the Terra+Aqua FBCT data between January 2003 and March 2022 with the NOAA-20 FBCT data for the remainder of the currently available record (from April 2022 to July 2024). Following ref. 35, we apply an adjustment to the unobscured low-cloud

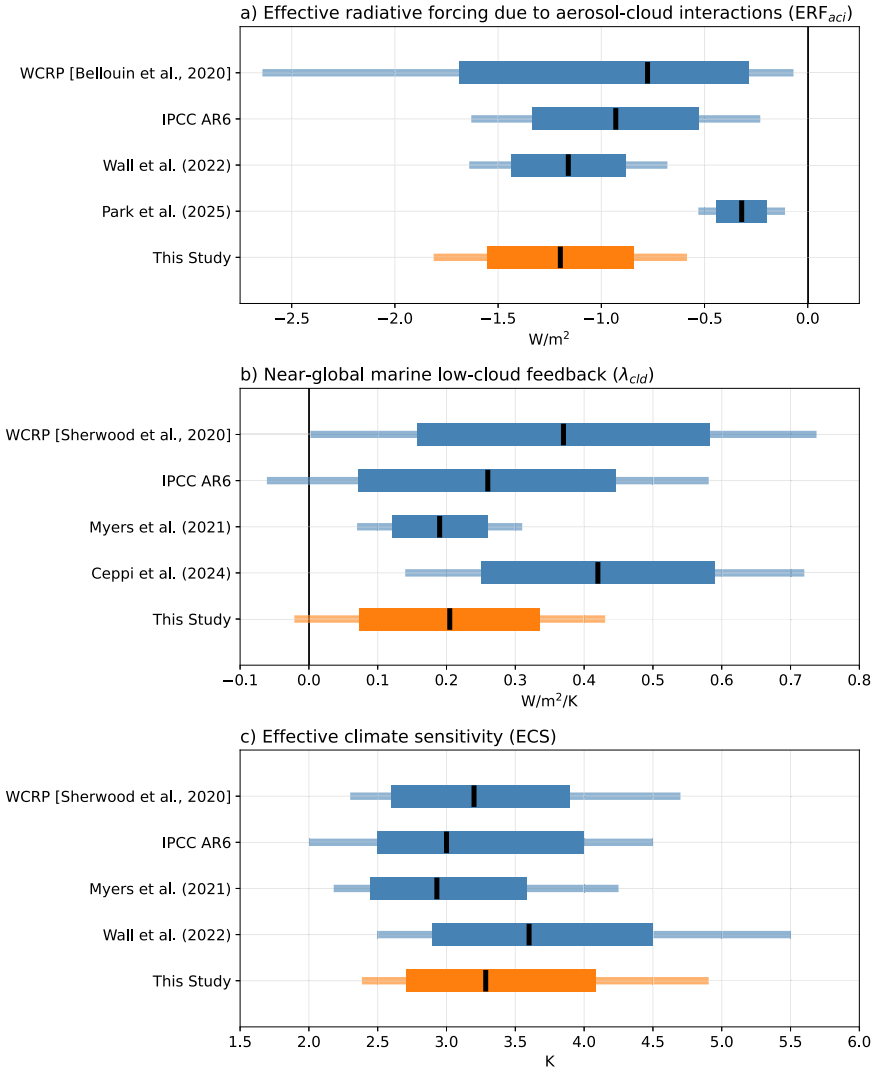


Fig. 4 | Estimates of aerosol indirect effect, low-cloud feedback, and climate sensitivity from this study in the context of previous analyses. Estimates of (**a**) global mean ERF_{aci}, (**b**) near-global marine low-cloud feedback, and (**c**) equilibrium climate sensitivity from previous studies and as estimated in this study. Thin and thick lines span the 90% and 66% confidence intervals, respectively, while the vertical line indicates the central estimates.

radiative effects from NOAA-20 FBCT such that the climatology over the period May 2018 - March 2022 matches the climatology from the Terra +Aqua FBCT dataset.

We consider a set of seven low-cloud controlling factors in this work: sea surface temperature (SST), estimated inversion strength (EIS), surface temperature advection (Tadv), 700 hPa relative humidity (RH_{700}), 700 hPa pressure velocity (ω_{700}), 10-meter wind speed (WS), and the natural logarithm of sulfate aerosol optical depth at 550 nm (ln(SAOD)). The first six predictors represent meteorological fields that have been shown to govern low cloud cover and have been used previously to constrain low cloud feedback[19,20,22]. These CCFs are computed from European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis v5 (ERA5) fields[36], or taken directly from ERA5 (in the case of SST, RH_{700}, ω_{700}, and WS). EIS is computed following ref. 37 using surface air temperature, surface pressure, and 700 hPa temperature and geopotential height. Tadv is computed from the product of 10 meter zonal and meridional winds and the corresponding zonal and meridional surface skin temperature gradient.

The addition of a seventh CCF to capture the aerosol influence on cloud properties follows ref. 21, but we use sulfate aerosol optical depth at 550 nm (SAOD) rather than sulfate aerosol concentration in the boundary layer. SAOD is taken from the ECMWF Atmospheric Composition Reanalysis 4 (EAC4), the fourth generation ECMWF global reanalysis of atmospheric composition provided by the Copernicus Atmosphere Monitoring Service (CAMS)[38,39]. To generate this product, satellite AOD retrievals from MODIS Dark Target product at 550 nm[40,41] and the Advanced Along-Track Scanning Radiometer[42] are assimilated into ECMWF's Integrated Forecasting System. We assessed the sensitivity of our results to several alternative aerosol predictors: total aerosol optical depth AOD from CAMS, aerosol index computed as the product of the Angstrom exponent and the aerosol optical depth at 550 nm from MODIS, and the 910 hPa sulfate aerosol concentration from MERRA2. All of these showed worse out-of-sample validation than the SAOD predictor (not shown).

We analyze data between January 2003 and December 2024, representing the period in which complete years are currently available across all datasets (except CERES-FBCT, which ends in July 2024). Anomalies are computed with respect to a monthly-resolved climatology computed over a January 2003 to December 2014 baseline period. All CCF anomalies are normalized to unit variance by dividing them by their respective interannual standard deviations over the baseline period. Prior to performing multilinear regression (next section), all predictors and predictands are regridded to 5° by 5° boxes to account for the adjustment time of clouds to their environment[19,20]. Data over land is excluded from all analyses.

CCF Analysis

R_{cld} sensitivities to CCFs are computed via multilinear regression as described below. This approach has been employed in several previous studies[19,30,31,43], and its validity, inherent assumptions, and limitations are discussed in a review article on the topic[44]. A key consideration when performing multilinear regression is whether the various predictor fields are sufficiently independent from each other. When CCFs are highly correlated

with one another, they contribute little additional predictive power to the regression model and are effectively redundant. Under such pronounced multicollinearity, estimates of R_{cld} sensitivity to individual predictors can become unstable, with small changes to the model or dataset leading to large and potentially misleading shifts in inferred predictor effects. Variance inflation factors (VIFs) provide a measure of this by quantifying the extent to which the variance of a regression coefficient is increased relative to a scenario in which the predictor is uncorrelated with others in the model. VIFs greater than 10 are commonly taken as evidence of substantial multicollinearity[45], though a lower threshold of 5 is sometimes used. In Supplementary Fig. 3. we show VIFs for each CCF. With the exception of the Eastern Pacific cold tongue region, where SST and EIS are highly correlated, and a few isolated pixels elsewhere, VIF values are generally below 5. This indicates that multicollinearity is not a problem for our analysis, especially since we primarily focus on the North Pacific and North Atlantic, where all locations have VIFs below 5.

We perform our regression analysis iteratively over each year in the record as follows: Both CCF_i and R_{cld} anomalies are first detrended. For a given target year, we remove that year and 2 years on either side of it from the dataset (a total of 5 years surrounding the target year). For years near the end of the record, we remove the available 5 adjacent years (e.g., for 2023, we remove 2020–2024). Using this subset of CCF_i and R_{cld} data with a 5-year period withheld, we perform ordinary least-squares multilinear regression to derive $\partial R_{cld}/\partial CCF_i$, which quantifies the sensitivities of R_{cld} to each CCF_i holding the other CCFs constant.

This process is done iteratively over each complete year in the data record (2003–2024), such that coefficients that will be used in the prediction of a given year are derived without "knowledge" of a 5-year period encompassing the target year or of the underlying trend in the data. Coefficient maps are qualitatively similar across target years (in each case withholding different 5-year windows), indicating that the sensitivities are largely invariant with time, an important prerequisite for CCF analysis. Results are also largely insensitive to the withholding window length, but coefficient uncertainties increase as more years are withheld. Note that the raw anomalies including the trend are used when making ΔR_{cld} predictions with these coefficients; only the dataset used to fit the multilinear regression model is detrended. This ensures that predictions of individual years and of the trend are out-of-sample tests.

The variance in R_{cld} explained by each CCF, computed as the linear correlation coefficient (squared) between each of the 7 CCF predictors and the R_{cld} predictand at each location across all months in the Jan 2003–Dec 2024 period, is shown in Supplementary Fig. 4a–g. The total variance explained by the multi-linear regression model is computed similarly as the linear correlation coefficient (squared) between the sum of all 7 CCF predictors' contributions and R_{cld} (Supplementary Fig. 4h).

Park et al. (2025)[29] argue that variations in SAOD primarily impact R_{cld} through their influence on cloud droplet number concentration (N_d), and therefore the sensitivity of R_{cld} to SAOD should be estimated as the product of two coefficients: $(\partial R_{cld}/\partial ln(N_d)) * (\partial ln(N_d)/\partial ln(\text{SAOD}))$. While our default methodology does not explicitly account for N_d mediation, we perform alternative calculations in which the $\partial R_{cld}/\partial ln(\text{SAOD})$ coefficient is replaced with this product. This is done by repeating the multilinear regression calculations above twice, first replacing $ln(\text{SAOD})$ with $ln(N_d)$ as the 7th CCF to yield $\partial R_{cld}/\partial ln(N_d)$; and second replacing R_{cld} with $ln(N_d)$ as the predictand to yield $\partial N_d/\partial ln(\text{SAOD})$. The product of these two coefficients, which is an estimate of the N_d-mediated R_{cld} sensitivity to SAOD, is then used in place of the original $\partial R_{cld}/\partial ln(\text{SAOD})$ coefficient that ignores mediation by N_d. As shown in ref. 29 and as found here, $\partial ln(N_d)/\partial ln(\text{SAOD})$ is generally less than one, such that the sensitivity of R_{cld} to SAOD is weaker when accounting for mediation by N_d (Supplementary Fig. 10).

For our sensitivity tests, we use N_d estimates from combined Aqua and Terra MODIS retrievals at 3.7 microns that have been constructed by ref. 46 using three increasingly stringent sampling strategies: those of refs. 47,48, and 49, each of which is applied on top of the one listed prior in addition to other filters. While all N_d dataset options (including ignoring N_d mediation entirely) lead to reasonable out-of-sample predictions of R_{cld} via the CCF framework (not shown), the derived ERF_{aci} estimates are sensitive to this choice (Supplementary Fig. 11). Given the substantial uncertainty in retrieving N_d from space, the desire to avoid using an aerosol predictor and a cloud-radiative predictand derived from measurements from the same satellite (MODIS), the desire to avoid inflating errors from combining two uncertain sensitivities ($\partial R_{cld}/\partial ln(N_d)$ and $\partial ln(N_d)/\partial ln(\text{SAOD})$), and the fact that aerosols can affect low cloud properties independent of their influence on N_d, we have elected to report as our baseline results from the less complex CCF framework that ignores mediation by N_d. The overall muted sensitivity of the posterior ECS to how N_d mediation is handled (Supplementary Fig. 9) highlights the robustness of our conclusion that recent observations do not fundamentally change our constraints on climate sensitivity.

Observationally-constrained ERF_{aci} and λ_{cld}

To derive observationally-constrained estimates of ERF_{aci}, we first multiply the R_{cld} sensitivity to $ln(\text{SAOD})$ by the model-predicted $\Delta ln(\text{SAOD})$ over the historical period[21]. We take $\Delta ln(\text{SAOD})$ as the difference between the 2005 and 2014 period and the 1850–1859 period in historical simulations averaged across 24 CMIP6 models that provided the necessary output (Supplementary Table 1). This product represents an observationally-constrained near-present ERF_{aci} due to low-level clouds over the oceans given modeled changes in aerosol loading since the 1850s. Following ref. 21, we relate the 55°S–55°N area-weighted mean of this quantity to the global (land+ocean) mean ERF_{aci} due to all cloud types using a scale factor (γ) determined from a ensemble of 19 CMIP6 models analyzed in ref. 50 that conducted the piClim-control and piClim-aer experiments as part of the Radiative Forcing Model Intercomparison Project[51] (Supplementary Table 1). These are idealized atmosphere-only simulations intended for diagnosing ERF_{aci} with aerosol burdens set to their preindustrial levels in the former experiment and to year-2014 levels in the latter experiment. We derive a multi-model mean γ of 1.06, very similar to that derived in ref. 21. Uncertainty in the resulting ERF_{aci} estimate (discussed further below) represents the combined uncertainty from observational regression coefficients and inter-model differences in both $\Delta ln(\text{SAOD})$ and γ.

To derive observationally-constrained near-global marine low cloud feedback, we follow the same procedure as ref. 20. Namely, we multiply the six meteorological coefficients by the change in these six CCFs in abrupt-4xCO_2 simulations from an ensemble of 85 CMIP5 and CMIP6 models that provided the necessary diagnostics (Supplementary Table 1). We compute ΔCCFs as the difference between the CCFs from the first 20 and last 20 years of CO_2 quadrupling experiments, normalized by the change in global mean surface temperature between these epochs. This product represents an observationally-constrained λ_{cld} due to low-level clouds over the oceans given model-predicted changes in meteorology in response to quadrupled CO_2. While alternative statistical optimization techniques have been applied in the literature to derive constrained cloud feedbacks via down-weighting models with biased cloud sensitivities[52], here we include all available CMIP5 and CMIP6 models' CCF responses. To the extent that model-predicted changes in meteorology are biased owing to missing or poorly-represented processes, the resultant cloud feedbacks will also be biased. By including as large an ensemble of climate models' ΔCCF predictions as possible, we attempt to account for a broad range of plausible outcomes. We scale this product by 0.644, the fraction of the globe covered by 60°S–60°N oceans, to yield the contribution to the global low-cloud feedback. Uncertainty in the resulting near-global mean estimate (discussed further below) represents the combined uncertainty from observational regression coefficients and inter-model differences in ΔCCFs.

Uncertainty calculations

Uncertainties in regression coefficients are estimated following standard practices[19–21]. Namely, the 95% confidence intervals of the regression

coefficients at a given grid point are computed as

$$\frac{\partial R_{\mathrm{cld}}}{\partial x_i} \pm t_{\mathrm{crit}} \sqrt{\mathbf{C_{ii}}} \sqrt{\frac{N_{\mathrm{nom}}}{N_{\mathrm{eff}}}}, \quad (1)$$

where x_i are detrended and standardized anomalies in CCFs and t_{crit} is the critical value of the Student's t distribution at the 95% significance level. $\mathbf{C_{ii}}$ represents the diagonal components of the variance-covariance matrix of meteorological anomalies, given by $\mathbf{C} = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$, where $\mathbf{X}$ is a matrix with columns composed of x_i and σ^2 is the mean squared error of the regression model. $N_{\mathrm{nom}}/N_{\mathrm{eff}}$ is the ratio of the nominal to effective number of monthly R_{cld} samples at a given grid box, estimated as $(1 + r_t)/(1 - r_t)$, where r_t is the lag-1 temporal autocorrelation of cloud-radiative anomalies.

Confidence intervals on CCF-estimated R_{cld} anomalies or trends at each grid box are estimated as

$$\Delta R_{\mathrm{cld}} \pm t_{\mathrm{crit}} \sqrt{\Delta x^T \mathbf{C} \Delta x} \sqrt{\frac{N_{\mathrm{nom}}}{N_{\mathrm{eff}}}} = \Delta R_{\mathrm{cld}} \pm \delta, \quad (2)$$

where Δx is a vector containing anomalies or trends in CCFs and one. For meteorology-only CCF predictions, Δx is a 7 × 1 vector containing the 6 meteorological CCF anomalies or trends and one. For aerosol-only CCF predictions, Δx is a 2 × 1 vector containing the aerosol CCF anomaly or trend and one. Δx used to derive observationally-constrained cloud feedbacks is a 7 × 1 vector containing CMIP5 and CMIP6 ensemble mean temperature-mediated changes in meteorological CCFs and one. Δx used to derive observationally-constrained $\mathrm{ERF_{aci}}$ is a 2 × 1 vector containing the CMIP6 ensemble mean change in $ln($ SAOD) and one.

Confidence intervals on spatially-averaged CCF predictions (indicated by angle brackets) are computed as:

$$\langle \Delta R_{\mathrm{cld}} \rangle \pm \sqrt{\frac{\sum_{k=1}^{N^*_{\mathrm{nom}}} (\delta_k w_k)^2}{\left(\sum_{k=1}^{N^*_{\mathrm{nom}}} w_k\right)^2}} \sqrt{\frac{N^*_{\mathrm{nom}}}{N^*_{\mathrm{eff}}}} = \langle \Delta R_{\mathrm{cld}} \rangle \pm \Delta_{\mathrm{obs}}, \quad (3)$$

where δ_k is the uncertainty from Eq 2 in the kth grid box, w_k is the cosine of latitude, and N^*_{nom} and N^*_{eff} are the nominal and effective number of 5° x 5° grid boxes, respectively. N^*_{eff} is computed by applying the method of ref. 53 to R_{cld} for all oceanic grid boxes between 60 °S and 60°N.

Confidence intervals on observationally-constrained cloud feedback account for an additional source of uncertainty from inter-model differences in ΔCCFs. We compute observationally-constrained near-global low cloud feedbacks using ΔCCFs from each individual CMIP GCM. Δ_{model} is estimated as half the 5–95 percentile range of these feedback values across GCMs. The total 90% confidence interval on regionally-averaged observationally-constrained low cloud feedback is then given by

$$\lambda_{cld} \pm \mathrm{CI} = \langle \frac{\Delta R_{\mathrm{cld}}}{\Delta T_g} \rangle \pm \sqrt{\Delta^2_{\mathrm{obs}} + \Delta^2_{\mathrm{model}}}. \quad (4)$$

We estimate a 90% confidence interval on the observationally-constrained cloud feedback of 0.23 W m^{-2} K^{-1}, made up of contributions from Δ_{obs} of 0.09 W m^{-2} K^{-1} and from Δ_{model} of 0.21 W m^{-2} K^{-1}. Inter-model differences in the response of EIS to warming is by far the dominant contributor to Δ_{model}, representing more than 95% of the variance in this term. Note that the uncertainty arising from inter-model differences in how CCFs respond to 4xCO_2 is more than twice as large as that from uncertainty in observational regression coefficients. This highlights that the lack of agreement in model-predicted CCF changes provides an important limit on how tightly observations can constrain cloud feedbacks through this methodology. Moreover, the uncertainty in our observationally-constrained low cloud feedback is considerably larger than that derived using essentially the same methodology in ref. 20 owing to our usage of ΔCCFs predictions from a much larger ensemble of climate models (85 rather than 18). The inflated uncertainty when accounting for a broader range of CCF changes echoes the results of ref. 54, who highlight the substantial uncertainty arising from large-scale meteorological conditions, which are of the same order as those attributed to subgrid parameterizations.

Confidence intervals on global-mean $\mathrm{ERF_{aci}}$ account for observational aerosol regression coefficient uncertainty, inter-model differences in $\Delta ln($ SAOD), and inter-model differences in γ, the parameter that links 55°S–55°N mean marine low-cloud $\mathrm{ERF_{aci}}$ to global mean land+ocean total $\mathrm{ERF_{aci}}$. The first two are computed as noted above for Δ_{obs} and Δ_{model}, except using aerosol-only uncertainties (denoted with subscript "A" for aerosol). The γ uncertainty is derived similarly to Δ_{model}, as half the 5–95 percentile range of γ values across GCMs. The net $\mathrm{ERF_{aci}}$ uncertainty is thus given by

$$\mathrm{ERF_{aci}} \pm \mathrm{CI} = \langle \Delta R_{\mathrm{cld}} \rangle \pm \sqrt{\Delta^2_{\mathrm{A,obs}} + \Delta^2_{\mathrm{A,model}} + \Delta^2_{\gamma}}. \quad (5)$$

We estimate a 90% confidence interval on the observationally-constrained $\mathrm{ERF_{aci}}$ of 0.614 W m^{-2}, made up of contributions from $\Delta_{\mathrm{A,obs}}$ of 0.30 W m^{-2}, $\Delta_{\mathrm{A,model}}$ of 0.44 W m^{-2}, and Δ_{γ} of 0.30 W m^{-2}.

Data availability

MODIS-COSP is available at https://ladsweb.modaps.eosdis.nasa.gov/missions-and-measurements/products/MCD06COSP_M3_MODIS. Cloud radiative kernels are available at https://zenodo.org/records/17704420[55]. CERES-FBCT data is available at https://ceres.larc.nasa.gov/data/. ERA5 reanalysis is available at https://www.ecmwf.int/en/forecasts/dataset/ecmwf-reanalysis-v5. EAC4/CAMS reanalysis data is available at https://ads.atmosphere.copernicus.eu/datasets/cams-global-reanalysis-eac4-monthly. MODIS-derived N_d data are available at the Centre for Environmental Data Analysis at https://doi.org/10.5285/864a46cc65054008857ee5bb772a2a2b[56]. CMIP5 and CMIP6 models used in this study are from the Earth System Grid Federation at https://aims2.llnl.gov/search. The WCRP ECS code is available at https://doi.org/10.5281/zenodo.3945276[57].

Code availability

Code to reproduce the results of this study are available at https://doi.org/10.5281/zenodo.18896195[58].

Received: 27 October 2025; Accepted: 19 March 2026;
Published online: 31 March 2026

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Acknowledgements

The effort of M.Z., Y.Q., L.W.C., and S.P. was supported by the U.S. Department of Energy (DOE) Office of Science Biological and Environmental Research program Regional and Global Model Analysis program area and was performed under the auspices of the DOE by Lawrence Livermore National Laboratory under Contract DE-AC52-07NA27344. T.M. was supported by NOAA cooperative agreement NA22OAR4320151. The statements, findings, conclusions, and recommendations are those of the authors and do not necessarily reflect the views of NOAA or the U.S. Department of Commerce. P.L.M. and A.G. were supported as part of the Enabling Aerosol-cloud interactions at GLobal convection-permitting scalES (EAGLES) project (project no. 74358), funded by the U.S. DOE, Office of Science, Office of Biological and Environmental Research, Earth System Model Development (ESMD) and Regional & Global Model Analysis (RGMA) program areas. The Pacific Northwest National Laboratory is operated for the U.S. Department of Energy by the Battelle Memorial Institute under contract DE-AC05-76RL01830. C.W. is funded by the European Union (ERC, AC3S, project number 101156240). Views and opinions expressed are however those of the authors only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them. P.C. was supported by UK Research and Innovation (UKRI) under the UK government's Horizon Europe funding Guarantee (grant EP/Y036123/1). P.C. was additionally supported through UK Natural Environmental Research Council (NERC) grants NE/V012045/1 and NE/T006250/1. This research used resources of the National Energy Research Scientific Computing Center (NERSC), a Department of Energy User Facility, using NERSC award BER-ERCAP0033047. We acknowledge the World Climate Research Programme, which, through its Working Group on Coupled Modelling, coordinated and promoted CMIP. We thank the climate modeling groups for producing and making available their model output, the Earth System Grid Federation (ESGF) for archiving the data and providing access, and the multiple funding agencies who support CMIP and ESGF.

Author contributions

Conceptualization: M.Z., A.G. Methodology: M.Z., T.M., C.W., P.C.; Investigation: M.Z., L.C. Visualization: M.Z. Writing - original draft: M.Z. Writing - review & editing: M.Z., T.M., Y.Q., L.C., S.K., S.P., P.M., C.W., P.C., A.G.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s43247-026-03461-8.

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Peer review information *Communications Earth & Environment* thanks the anonymous reviewers for their contribution to the peer review of this work. Primary Handling Editors: Zijun Li, and ChenRui Diao. A peer review file is available

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